



# ANGLES

## UNEQUAL

| Section Size<br>A x B x t | Moment of Inertia |                 |                 |                 | Max Iu          |                 |                 |                 | Radius of Gyration |       |      |       |        |       |        |       | tan   | Modulus of Section |                 |                 |                 |
|---------------------------|-------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|--------------------|-------|------|-------|--------|-------|--------|-------|-------|--------------------|-----------------|-----------------|-----------------|
|                           | Ix                |                 | Iy              |                 |                 |                 | Min Iv          |                 | ix                 |       | iy   |       | Max Iu |       | Min Iv |       | a     | Zx                 |                 | Zy              |                 |
|                           | cm <sup>4</sup>   | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm <sup>4</sup> | in <sup>4</sup> | cm                 | in    | cm   | in    | cm     | in    | cm     | in    |       | cm <sup>3</sup>    | in <sup>3</sup> | cm <sup>3</sup> | in <sup>3</sup> |
| 40 x 20 x 3               | 2.81              | 0.068           | 0.47            | 0.011           | 2.98            | 0.072           | 0.31            | 0.007           | 1.27               | 0.50  | 0.52 | 0.205 | 1.31   | 0.516 | 0.42   | 0.165 | 0.257 | 1.09               | 0.067           | 0.30            | 0.018           |
| 40 x 20 x 5               | 4.21              | 0.11            | 0.68            | 0.016           | 4.43            | 0.106           | 0.41            | 0.01            | 1.24               | 0.488 | 0.50 | 0.197 | 1.27   | 0.50  | 0.41   | 0.161 | 0.243 | 1.68               | 0.103           | 0.46            | 0.208           |
| 50 x 35 x 4               | 8.08              | 0.194           | 3.25            | 0.078           | 9.49            | 0.228           | 1.84            | 0.044           | 1.57               | 0.618 | 0.99 | 0.39  | 1.70   | 0.669 | 0.75   | 0.295 | 0.475 | 2.36               | 0.144           | 1.23            | 0.075           |
| 50 x 35 x 6               | 12.2              | 0.269           | 4.43            | 0.106           | 13.1            | 0.315           | 2.55            | 0.061           | 1.54               | 0.606 | 0.97 | 0.382 | 1.66   | 0.654 | 0.73   | 0.287 | 0.466 | 3.35               | 0.204           | 1.72            | 0.105           |
| 60 x 50 x 5               | 18.5              | 0.445           | 11.7            | 0.281           | 24.4            | 0.586           | 5.82            | 0.14            | 1.87               | 0.736 | 1.48 | 0.583 | 2.14   | 0.843 | 1.05   | 0.413 | 0.679 | 4.40               | 0.269           | 3.16            | 0.193           |
| 60 x 50 x 7               | 24.4              | 0.586           | 15.2            | 0.365           | 31.9            | 0.766           | 7.68            | 0.185           | 1.84               | 0.724 | 1.45 | 0.571 | 2.10   | 0.827 | 1.03   | 0.406 | 0.673 | 5.89               | 0.359           | 4.2             | 0.256           |
| 65 x 50 x 5               | 23.2              | 0.557           | 11.9            | 0.286           | 28.8            | 0.692           | 6.33            | 0.152           | 2.04               | 0.803 | 1.47 | 0.579 | 2.28   | 0.898 | 1.07   | 0.421 | 0.577 | 5.14               | 0.314           | 3.19            | 0.195           |
| 65 x 50 x 7               | 30.6              | 0.735           | 15.6            | 0.375           | 37.9            | 0.911           | 8.35            | 0.201           | 2.01               | 0.791 | 1.44 | 0.567 | 2.24   | 0.882 | 1.05   | 0.413 | 0.571 | 6.90               | 0.421           | 4.25            | 0.259           |
| 65 x 50 x 9               | 37.8              | 0.908           | 19.2            | 0.461           | 46.6            | 1.12            | 10.4            | 0.25            | 1.99               | 0.783 | 1.42 | 0.559 | 2.21   | 0.87  | 1.05   | 0.413 | 0.565 | 8.69               | 0.53            | 5.33            | 0.325           |
| 70 x 60 x 6               | 35.3              | 0.848           | 23.9            | 0.574           | 47.6            | 1.144           | 11.7            | 0.281           | 2.17               | 0.854 | 1.78 | 0.701 | 2.15   | 0.988 | 1.24   | 0.488 | 0.719 | 7.17               | 0.438           | 5.4             | 0.33            |
| 70 x 60 x 8               | 44.4              | 1.067           | 29.8            | 0.716           | 59.5            | 1.429           | 14.7            | 0.353           | 2.13               | 0.839 | 1.75 | 0.689 | 2.47   | 0.872 | 1.23   | 0.484 | 0.714 | 9.13               | 0.457           | 6.85            | 0.418           |
| 70 x 60 x 10              | 53.6              | 1.288           | 36.0            | 0.865           | 71.7            | 1.723           | 17.9            | 0.43            | 2.11               | 0.831 | 1.73 | 0.681 | 2.44   | 0.961 | 1.22   | 0.48  | 0.711 | 11.2               | 0.683           | 8.41            | 0.513           |
| 75 x 50 x 6               | 40.4              | 0.971           | 14.4            | 0.346           | 46.4            | 1.115           | 8.33            | 0.20            | 2.36               | 0.929 | 1.41 | 0.555 | 2.53   | 0.996 | 1.07   | 0.421 | 0.434 | 7.97               | 0.486           | 3.78            | 0.231           |
| 75 x 50 x 8               | 50.8              | 1.221           | 17.8            | 0.428           | 58.1            | 1.396           | 10.4            | 0.25            | 2.33               | 0.917 | 1.38 | 0.543 | 2.49   | 0.98  | 1.05   | 0.413 | 0.427 | 10.2               | 0.622           | 4.77            | 0.291           |
| 75 x 50 x 10              | 61.4              | 1.475           | 21.4            | 0.514           | 70.0            | 1.682           | 12.7            | 0.257           | 2.31               | 0.909 | 1.36 | 0.535 | 2.47   | 0.972 | 1.05   | 0.413 | 0.421 | 12.5               | 0.763           | 5.85            | 0.357           |
| 75 x 65 x 6               | 44.1              | 1.06            | 30.7            | 0.738           | 49.9            | 1.439           | 14.8            | 0.356           | 2.33               | 0.917 | 1.94 | 0.764 | 2.72   | 1.071 | 1.35   | 0.531 | 0.737 | 8.3                | 0.506           | 6.39            | 0.523           |
| 75 x 65 x 8               | 55.5              | 1.333           | 38.4            | 0.923           | 75.3            | 1.809           | 18.6            | 0.447           | 2.29               | 0.902 | 1.91 | 0.752 | 2.67   | 1.051 | 1.33   | 0.524 | 0.733 | 10.6               | 0.647           | 8.13            | 0.496           |
| 75 x 65 x 10              | 67.3              | 1.617           | 46.5            | 1.117           | 91.0            | 2.186           | 22.8            | 0.548           | 2.27               | 0.894 | 1.89 | 0.744 | 2.65   | 1.043 | 1.32   | 0.52  | 0.73  | 13.0               | 0.793           | 9.99            | 0.61            |
| 80 x 60 x 6               | 51.5              | 1.237           | 24.8            | 0.596           | 62.9            | 1.511           | 13.4            | 0.322           | 2.52               | 0.992 | 1.75 | 0.689 | 2.78   | 1.094 | 1.28   | 0.504 | 0.548 | 9.30               | 0.568           | 5.5             | 0.336           |
| 80 x 60 x 8               | 65.0              | 1.562           | 31.0            | 0.745           | 79.1            | 1.90            | 16.9            | 0.406           | 2.48               | 0.976 | 1.71 | 0.673 | 2.74   | 1.079 | 1.26   | 0.496 | 0.542 | 11.9               | 0.726           | 6.97            | 0.425           |
| 80 x 60 x 10              | 78.7              | 1.891           | 37.4            | 0.899           | 95.6            | 2.297           | 20.5            | 0.493           | 2.46               | 0.969 | 1.7  | 0.669 | 2.52   | 0.992 | 1.26   | 0.496 | 0.538 | 14.6               | 0.891           | 8.55            | 0.522           |
| 80 x 70 x 6               | 54.1              | 1.30            | 38.6            | 0.927           | 75.3            | 1.785           | 18.4            | 0.442           | 2.49               | 0.98  | 2.1  | 0.827 | 2.92   | 1.15  | 1.45   | 0.571 | 0.752 | 9.52               | 0.581           | 7.46            | 0.455           |
| 80 x 70 x 9               | 75.8              | 1.821           | 53.8            | 1.293           | 104             | 2.499           | 25.9            | 0.622           | 2.44               | 0.961 | 2.06 | 0.811 | 2.86   | 1.126 | 1.43   | 0.563 | 0.748 | 13.6               | 0.83            | 10.6            | 0.647           |
| 80 x 70 x 12              | 96.6              | 1.321           | 68.3            | 1.641           | 131             | 3.147           | 33.4            | 0.802           | 2.41               | 0.949 | 2.03 | 0.799 | 2.82   | 1.11  | 1.42   | 0.559 | 0.744 | 17.7               | 1.08            | 13.8            | 0.842           |
| 90 x 60 x 6               | 71.5              | 1.718           | 23.6            | 0.615           | 82.3            | 1.977           | 14.8            | 0.356           | 2.86               | 1.126 | 1.71 | 0.674 | 3.07   | 1.209 | 1.30   | 0.512 | 0.437 | 11.7               | 0.714           | 5.47            | 0.34            |
| 90 x 60 x 9               | 101               | 2.427           | 35.4            | 0.85            | 115             | 2.763           | 20.7            | 0.497           | 2.82               | 1.11  | 1.67 | 0.657 | 3.01   | 1.185 | 1.28   | 0.504 | 0.43  | 16.7               | 1.091           | 7.9             | 0.482           |
| 90 x 60 x 12              | 128               | 3.075           | 44.8            | 1.076           | 146             | 2.508           | 26.7            | 0.641           | 2.78               | 1.094 | 1.65 | 0.65  | 2.97   | 1.169 | 1.27   | 0.50  | 0.423 | 21.8               | 1.33            | 10.2            | 0.622           |
| 90 x 75 x 6               | 76.9              | 1.848           | 48.6            | 1.168           | 101             | 2.427           | 24.2            | 0.581           | 2.83               | 1.114 | 2.25 | 0.886 | 3.25   | 1.28  | 1.58   | 0.622 | 0.681 | 12.1               | 0.738           | 8.68            | 0.53            |
| 90 x 75 x 9               | 109               | 2.619           | 68.1            | 1.636           | 143             | 3.436           | 34.1            | 0.819           | 2.78               | 1.094 | 2.2  | 0.866 | 3.19   | 1.257 | 1.56   | 0.614 | 0.676 | 17.4               | 1.062           | 12.4            | 0.757           |
| 90 x 75 x 12              | 139               | 3.34            | 86.8            | 2.085           | 182             | 4.373           | 44.0            | 1.057           | 2.75               | 1.083 | 2.17 | 0.854 | 3.14   | 1.236 | 1.55   | 0.61  | 0.672 | 22.6               | 1.379           | 16.1            | 0.982           |